How ChatGPT can be used for fashion

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1 Introduction

The gradual integration of Artificial Intelligence (AI) into the fashion and textile industry, spanning from 2013 to 2019, marked a period of unprecedented transformation. Initially, AI was primarily focused on personalizing product recommendations, leveraging machine learning algorithms to analyze consumer purchase histories, individual preferences, and even social media interactions. This initial wave of innovations allowed fashion companies to offer more precise product suggestions, thereby enhancing the online shopping experience. [8]

Simultaneously, AI made inroads into the operational management of the industry, optimizing manufacturing processes and the supply chain. AI-based systems were deployed to predict demand, manage inventory levels more effectively, and automate specific production steps. This operational optimization contributed to cost reduction and improved efficiency across the industry. [5]

During this period, the emergence of computer vision and natural language processing (NLP) represented a significant turning point. Computer vision facilitated the visual analysis of fashion trends, based on image recognition from fashion shows, street styles, and various visual influences. NLP, on the other hand, was utilized to enhance understanding of customer needs by analyzing social media comments, online reviews, and other textual data.

This technological convergence also led to generative models, such as Generative Adversarial Networks (GANs). These models introduced a creative dimension by enabling the computer-generated design of clothing, exploring new frontiers of creativity based on patterns and styles learned from

extensive datasets.

However, even with these exciting advancements, it is imperative to acknowledge that this revolution is not without controversies. The increasing reliance on AI raises significant ethical questions, including concerns about consumer privacy and algorithm transparency. Furthermore, while generative models stimulate creativity, debates are emerging regarding the role of AI in the creative process and its impact on human artistic authenticity.

As AI redefines the fashion industry, it is crucial to adopt a critical perspective, exploring opportunities while addressing challenges to ensure an ethical and balanced integration of this fusion between technology and creativity.

2 What is Artificial Intelligence?

2.1 Definition of Artificial Intelligence

Artificial Intelligence is a field of computer science focused on developing computer systems capable of performing tasks that typically require human intelligence. AI aims to create systems capable of reasoning, learning from experience, adapting to new situations, and functioning autonomously.

According to Marvin Lee Minsky, artificial intelligence AI is defined as "the construction of computer programs that engage in tasks currently performed more satisfactorily by human beings, as they require high-level mental processes such as perceptual learning, memory organization, and critical reasoning." [29]

2.2 History of AI

In the 1940s and 1950s, Alan Turing played a crucial role in developing theoretical computer science. He invented a mathematical model of computation "Turing machine".

The term of artificial intelligence was used by John McCarthy, one of the pioneers in this field, during the "Dartmouth Summer Research Project on Artificial Intelligence" in 1956. This summer workshop is considered as a founder of AI.
After this event, many researchers worked on the subject and developed programs that can resolve simple problems and play games like chess.

Despite the great enthusiasm, the period between 1970 and 1980 was called AI winter due to the technical limits especially the lack of computa-

tional power. In fact, computers were not able to store enough information or process it fast enough. Funding has decreased and public interest too. Nevertheless, research in specialized areas continued such as logic programming, neural networks, and computer vision.

The late 20th century marked the renewed interest in artificial intelligence. Many advancements have been made especially in data availability and computational power. In 1989, Yann Lecun developed the first neural network capable of recognizing handwritten numbers, which will give rise to deep learning. Also, a major event marks the history of AI, IBM's Deep Blue system triumphs over world chess champion Garry Kasparov. For the first time, the machine has defeated man.
Nowadays, the use of artificial intelligence raises significant ethical, social, and economic issues.
[32]
[4]

2.3 Different types of AI

2.3.1 General Artificial Intelligence

General AI or deep AI can perform any cognitive task like a human being or an animal. AI researchers believe that humanity possesses the technology necessary to create general AI, mainly thanks to advances in neural networks.

2.3.2 Weak Artificial Intelligence

Weak AI or specialized AI can perform a specific task without demonstrating learning capacities. This type of AI is very efficient in their specific field but cannot generalize their skills beyond that. Weak AI can be used in various fields such as facial recognition, voice recognition, or GPS navigation systems. Unlike strong AI, which has intelligence equivalent to or greater than human beings, weak AI does not allow reproduce or simulate human intelligence.

2.3.3 Strong AI

Strong AI refers to an artificial intelligence that can show signs of a clean conscience. This form of AI could be able to learn, understand, and apply knowledge like a human and have the capacity for independent thought. According to AI researchers, this type of AI cannot be created now because the concept of consciousness cannot be integrated into a mathematical system.

Moreover, strong AI can raise philosophical and ethical questions about the impact on our society and the relationship between humans and machines. Even if strong AI remains a theoretical concept, it continues to be a debate for scientists [29].

2.4 AI Techniques

2.4.1 Machine learning

Supervised learning is a machine learning technique that allows an algorithm to train from labelled data. These data include pairs of inputs and their corresponding expected outputs. The algorithm will adjust parameters to minimize the error between predicted outputs and actual outputs. Once trained, the algorithm can be used to predict outputs for new data based on the information provided during training. However, it can be difficult to collect data with the expected answers. Supervised learning is one of the most common approaches in machine learning. It can be used for speech recognition, automatic translation, or product recommendations.

Unsupervised learning is a machine learning technique where an algorithm can be trained from unlabeled data. During training, there are no output labels provided. The algorithm has to find links and structures in the data. One of the most popular applications is clustering: the algorithm combines similar data points according to their characteristics. Otherwise, another application can be anomaly detection: the algorithm identifies data points that can be put together.

Reinforcement learning is a machine learning technique where an intelligent agent (the agent who can act intelligently) can interact with an environment and learn to make decisions, good or bad, and depending on his action can have rewards or penalties. The agent's objective is to learn strategies to optimize rewards over time. This technique is often used when the agent's actions directly influence its environment such as video games or robotics.

[2]

2.4.2 Artificial neural networks

Artificial neural networks are inspired by the functioning of the human brain and are organized into layers: input layer, hidden layers, and output layer. Each neuron receives input signals, performs a computation, and applies an activation function to produce an output, transmitted to the neurons in the next layer. Neural networks rely on training data to learn and improve their accuracy over time. Once these learning algorithms are sufficiently refined,

they become powerful tools for computing and artificial intelligence. They can quickly classify and group data. Tasks such as speech or image recognition can be executed in a few minutes. One of the most famous neural networks is Google's search algorithm.

[1]

2.4.3 Deep learning

Deep learning can analyze unstructured data such as images, videos, text... It uses several layers of neurons. The more layers, the deeper the learning. This depth allows the network to learn hierarchical representations of the data, extracting complex features from simpler features. This model works well with a large amount of data. Deep learning is used in different fields such as image and speech recognition, natural language processing, healthcare, finance, autonomous car...

[36]

3 The influence of Artificial Intelligence in the Textile Industry and its consequences

3.1 Impact of AI in the textile and fashion industry

3.1.1 Enhancement of design and creation processes through AI

Improving design and creation processes through AI in the textile industry represents a significant evolution that essentially changes how products are conceived and developed.

Different aspects of this improvement encompass a wide spectrum of benefits :

- Advanced Creativity: Suggested ideas, innovative ideas, and novel complements to concepts are assured through AI for designers. Based on sophisticated data analysis, AI identifies emerging trends, repeating motifs, and consumer tendencies to provide a firm basis for creative inspiration.

- Streamlining Design Processes: AI streamlines design workflows by automating repetitive and arduous tasks such as pattern creation and design modification, which frees up designers for more complex and creative aspects of their work.

- Reduced Development Time: By automating some design steps and improving workflow efficiency, AI cuts down product development time con-

siderably. This makes companies agile enough to adapt quickly to changes in the market and roll out products faster than others, thus winning the game.

[30]

3.1.2 AI serving personalization of product recommendations and customer experience- Virtual clothing try-on by Max Planck Institute

AI facilitate the provision of products that are more personalized to each customer's unique preferences. Many machine learning algorithms can analyze customer data from social media interactions to purchase history and suggest products that correspond to their known tastes and concerns.

The AI algorithms analyze the purchase history of a client, their past interactions on your website, declared preferences, and contextual data such as weather, time of day. AI systems monitor the client's reaction and action to the recommendations, which helps in the extension of data about this particular user and the fine-tuning of the offer. Artificial intelligence can also forecast a client's future actions based on their behavior in the past and foreseen patterns. [16]

Virtual clothing try-on: A virtual clothing try-on is technology that utilizes 3D models to help consumers to see how a specific garment would look like on their body. Rather than simply displaying static images of the garment, this technology uses virtual avatars that can be adjusted to match the customer's measurements. This will allow consumers to see how the garment would actually fit their real body shape and size.

The virtual clothing try-on process typically involves the following steps:
- Capture of Anthropometric Data: The customer gives precise body measurements, such as their weight, height,measurements, and other details.
- Creation of a Virtual Avatar: thanks to those data, a 3D model representing the customer's body can be generated.
- Trying on Clothes: Then, the customer can select specific clothes to try on. These clothes are then adjusted on the virtual avatar to reflect their actual cut and size.
- Visualization and Customization: The customer can see the avatar wearing the selected clothes from different angles. They can also customize some aspects, like the colour or style of the garment.

This improves the online shopping experience and reduces product returns. [11]

The Max Planck Institute for Informatics is a research institute based in Germany focusing on various aspects of computer science, including AI.

[13]

3.1.3 Market Trend Anticipation- Example of Rita Sleiman

Rita Sleiman is an essential figure in anticipating market trends in fashion industry. She is known for her expertise in data analysis and her ability to precisely predict emerging trends that will influence consumer choices.
To anticipate market trends, she generally uses a combination of traditional methods and modern techniques, some of which are based on artificial intelligence and machine learning.

Different approaches include:
Data Analysis: The collection and analysis of data from a wide range of sources, including social media, e-commerce platforms, fashion blogs, magazines, or fashion shows, will put her in a position to trace emerging trends, repeating patterns, and changes in consumers' tastes.

• Influencer Monitoring: Influencers play a significant role in spreading fashion trends. Rita Sleiman closely monitors the activities of key influencers on social media to identify styles and products gaining popularity.
• Use of AI and Machine Learning: Rita Sleiman uses machine learning algorithms to analyze data and identify hidden trends. These algorithms can detect complex patterns that would remain inconspicuous to human analysts in data.
• Qualitative Analysis: She also considers qualitative factors like cultural events, technological developments, and societal changes that could influence fashion trends.
•Long-term Forecasting: Other than immediate trends, she would also forecast the future development of the fashion industry in the long run to give strategic insight to the brands regarding the design, production, and marketing of the products. [33]

3.2 Ethical reflection on human creativity and customer choice

3.2.1 Ethical implications of AI use in artistic creation and fashion

There is a serious challenge to ethical questions and debates regarding the use of AI in the creation of art and fashion.
• Originality and Creativity: Now, whether AI can actually be creative in the human sense of the word is a hotly debated subject. Some argue that AI can only work on pre-existent models and training data to produce its work, hence questioning the level of real originality and innovation in something that AI has created. This raises questions about the artistic and cultural value of such works in comparison to those created by human artists.
• Responsibility and Intellectual Property: In that case of creation by AI, it becomes a challenge to know who owns the responsibility for works and who

holds the copyright. Intellectual property and artistic credit issues might become blurred, especially when there is the involvement of multiple entities in the process of creation. [24]

• Impact on Artists and Creators: Widespread adoption of AI in artistic creation and fashion may also bring with it serious implications for the livelihoods of human artists and creators. This would further push traditional artists to the periphery if AI-generated works become more popular or preferred than those created by humans.

Ethical concerns associated with AI-generated art include potential copyright violations and moral issues related to scraping photos without licenses.

[22]

3.2.2 Critical thinking regarding personalization and AI influence on purchasing decisions

During personalized shopping experiences, AI brings about a particular manipulation of choices: AI systems can be designed to subtly influence customers' purchasing decisions by highlighting certain products or using persuasion techniques such as limiting offers or artificial urgency. This can lead consumers to make impulsive decisions or purchase products they don't actually need.

To personalize product recommendations, companies are often collecting and analyzing large amounts of customers' data. This raises concerns about privacy and data security, especially if this information is used for purposes other than personalizing shopping experiences.

Furthermore, with personalized product recommendations, consumers may develop an increased dependency on online platforms for their purchases, which can reduce the diversity of information sources and limit opportunities for price and product comparisons.

Finally, through its subtle impact on purchasing decisions, AI makes consumers lose their free will, as they find themselves buying items they probably would not have chosen if it weren't for the AI-driven influences.

[18]

4 Evolution of AI in Fashion

Incorporating artificial intelligence into design and modelling processes is revolutionizing the fashion industry, opening up new opportunities for designers. It has revolutionized many aspects of our society, and the fashion industry is no exception.

AI has been an integral part of how fashion companies conceive, produce, and market clothing. Various image recognition, predictive analysis, and content generation technologies have allowed fashion brands to further

creative boundaries and predict market behavior while also offering consumers a more personalized experience.

4.1 Initial use of AI in design and modelling processes

In this context, Computer-Aided Design (CAD) and 3D modelling software play a central role, merging human creativity with the analytical power of algorithms to shape the future of fashion.

1. Adoption of Computer-Aided Design (CAD) and 3D modelling software

Meanwhile, AI allows CAD to reach entirely new heights. In particular, while processing the archived history of stylistic tropes, this software features machine learning algorithms that navigate designers in the process of creating new "style moves". In fact, this is a territory of not only shapes, patterns, and even colour combinations that statistically should soon become popular. This approach allows the AI CAD to decide on the designs faster and extend its imagination beyond the conventional design paradigms. Thus, CAD becomes an even more formidable land more with the power of AI. [10]

At the same time, 3D modelling software is also transforming the way designers think and showcase their products. In fact, in combination with other software, for example AI, it can be used to generate a virtual avatar who may put on digital prototypes of dissemination. In this way, it becomes possible to understand how the model appears in three dimensions, which allows the designers to make decisions faster and eliminate the need for expensive physical prototypes. This interactive method also allows fit-review, which in turn enhances the design accuracy and production resources planning. [6]

The blending of CAD, 3D modelling software and AI recasts the fashion terrain, as the creative visions of designers and the prescience of algorithms converge into a mutual partnership. This partnership is the crucible for creativity and efficiency, and it defines a future where designs of unparalleled splendour emerge from the integration of human and artificial creativity.

4.2 Use of image recognition

4.2.1 Rapid Sorting of Millions of Products for Trend Monitoring

Image recognition powered by artificial intelligence revolutionizes how fashion businesses monitor and respond to emerging trends. Using sophisticated algorithms, this technology enables fast and efficient sorting of millions of

products from various sources such as social media, fashion shows, or online shopping sites. This capability of rapid and automated sorting provides brands with the ability to track market developments in real-time and instantly respond to changes in consumer preferences. [16]

4.2.2 Rapid Identification of Emerging Styles in the Market

Another opportunity, enabled by image recognition, is the ability to rapidly detect new and upcoming styles on the market. For instance, if the AI is scanning pictures of the most popular products or products that are becoming popular, it can detect new patterns, cutting designs, or the most fashionable colors. By viewing waves of consumer preferences at the moment they emerge, brands can integrate these elements into their offer in record time. For example, if a clothes brand sees a wave of popularity for a particular type of pattern or silhouette, it can quickly include it in the new collection. [16]

4.3 Predictive analysis for demand forecasting

4.3.1 Forecasting Demand Evolution Based on AI

In general, integrating predictive analysis into stock management and production planning operations presents a considerable advantage for fashion companies. By deploying advanced machine learning models, AI has access to a broad scope of data, from sales histories and seasonal trends to customer behaviors or the impact of external variables like the weather or social events. Thus, companies can predict market demand fluctuations with a higher degree of accuracy, enable their production to adjust to that estimate and optimize their inventory for efficient market demand servicing. Namely, a clothing company would be able to adjust the level of production according to the season forecasts and, hence, would never over- or understock. [26]

4.3.2 Refinement of Pricing and Inventory Management Strategies

Moreover, AI-based predictive analysis can be used to enhance pricing and inventory management and identify opportunities to generate revenue without incurring costs due to unsold goods or stockouts. The use of AI allows businesses to analyze demand dynamics, the reaction of consumers to price changes, and seasonality trends to identify optimal price and adjust their inventory. In this way, it is possible to develop various pricing strategies,

including offering a discount on a good that is likely to be overstocked, increasing the price of product in high demand, etc.

4.4 Introduction of Generative Adversarial Networks

4.4.1 AI-assisted Artistic Creation and Visual Generation

Fashion creators can also find a new dimension to their art thanks to the integration of Generative Adversarial Networks into the design process. This type of algorithms can produce realistic images when fed random input data, which gives designers an opportunity to produce styles, patterns, and textures indefinitely. Thus, the use of GANs helps creators remain constantly inspired in a field offering an infinite amount of design choices that have already been utilized; with the endless amount of choices, a designer can generate a new pattern or use GANs to create a digital masterpiece, which can then be implemented into an actual form. [37]

4.5 Source of Inspiration and Unprecedented Creativity for Designers

Relying on GANs results in new image creation in addition to the generation of the generated images; they stimulate creativity by generating new ideas and concepts. Designers, using the capabilities of GANs, are able to think over traditional decisions, introduce something unique, and innovative. For instance, they can create the pattern of virtual fabric similar to natural texture or suggest style preferences based on the analysis of what was popular before and what is popular now. Subsequently, the partnership between AI and field-specific creativity contributes to the growth of innovations in the fashion field and the possibility to present more collection that astonishes the public.

5 Advanced Supply Chain Management through AI

5.1 Predictive analysis for demand forecasting

5.1.1 Analysis of historical data for demand forecasting

Using machine learning algorithms that analyze historical demand trends to predict future needs started several years ago. However, these techniques have become popular in recent years with the rise of AI and big data.

These techniques have wide usage in the domain of supply chain management for the estimation of demand for different products and raw materials. For example, manufacturing industries use predictive models to estimate the demand for finished products based on seasonal variations, market trends, and marketing campaigns.

In the distribution realm, retailers utilize machine learning algorithms to anticipate demand for certain products in various geographies, taking into consideration variables as local as the weather, local events, and customer preferences.

Using predictive analytics tools within the fashion industry enables one to anticipate fashion trends and predict demand for certain styles, colors, or types of clothing—like many online fashion companies, SHEIN collects a large amount of data on customer preference, past sales, social media interaction, and industry trends. These predictions will help manufacturers and retailers adjust their production and inventory according to the wishes of the consumers, hence reducing the chances of overstocking or shortages.

Some other common tools in the fashion industry used for such analyses include predictive analytics software like SAS Fashion Forecasting, IBM Watson Trend, and data analytics platforms like Tableau and Power BI. These tools integrate machine learning algorithms and advanced analytics techniques to process huge amounts of data and generate accurate demand forecasts.

5.1.2 Optimization of stock levels and resource utilization

By incorporating these predictive models, companies can effectively fine-tune their inventory management strategies to shifting consumer preferences in ways that yield efficiency and resilience in their supply chain management practices.

In the domain of inventory management, firms use predictive models to determine optimal stock levels to enhance operational efficiency and control costs. A commonly followed approach is the Just-In-Time inventory management approach aimed at keeping minimal stock levels through synchronization of supplies with actual demand. With JIT, companies minimize inventory holding costs by avoiding excessive storage while ensuring adequate product availability to meet customer demand.

Another technique used to classify items by their relative value is known as ABC-based inventory management. The ABC method categorizes items

under three main categories: A, B, and C. Items falling under category A represent those that are of a high value but low volume. Items in category C represent those that are of low value but high volume. By classifying items in this manner, companies are able to focus their efforts on managing the most critical items in the inventory—category A—while adopting less rigorous inventory management policies for less critical items—category C.

Integrating artificial intelligence into virtual try-ons with the use of interactive mirrors is a great advantage for stock optimization, whereby the risk of returns from customers is reduced compared to ordering online.

By integrating these techniques into their inventory management process, companies will be able to achieve an optimal balance among customer satisfaction, cost reduction, and operational efficiency.

5.2 Optimized route planning and transportation

5.2.1 Route optimization considering traffic, energy efficiency, and delivery schedules

That is route optimization and transport to use advanced algorithms to determine the most effective vehicle routes, considering real-time traffic conditions, energy efficiency considerations, and delivery schedules. Through dynamic route adjustment and traffic pattern analysis, companies would be able to reduce transportation time and fuel consumption, while at the same time keeping goods on time for delivery. Therefore, the operation costs can be minimized with resources optimized, reducing transportation costs and increasing the economic efficiency of the supply chain.

5.2.2 Reduction of fuel consumption, greenhouse gas emissions, and costs by choosing sustainable suppliers

For such purposes, companies tend to apply sustainable logistics techniques: shipment consolidation, cargo consolidation, and optimal utilization of vehicle loading capacities. All these approaches, if prioritized by any company, help reduce fuel consumption, greenhouse gas emissions, and transportation costs while contributing to environmental preservation and a more sustainable supply chain.

5.3 Visibility and transparency in the supply chain

Real-time improvement in the visibility of movements and transactions

Real-time improvement in the visibility of movements and transactions Companies apply AI-powered technologies, such as real-time tracking sys-

tems and blockchain, which allow them to give a complete view of goods movement and transactions at each step in the supply chain. These technologies enable real-time access to data on inventory levels, shipping statuses, and production processes for stakeholders, making it easier to make decisions and manage risks. Moreover, by bringing transparency into processes and transactions, AI enables trust to be built among partners and stakeholders, while not causing any breaches in compliance with regulatory requirements. Overall, integrating AI-driven visibility and transparency solutions optimizes supply chain operations, reduces risks, and improves overall efficiency and accountability.

5.4 Traceability

5.4.1 Detection of anomalies in production stages

It allows for meticulous tracking of quality and quantity at every stage of production. This enables the early detection of anomalies or deviation from expected standards, thus ensuring that high-quality products are consistently maintained. By enhancing accountability among the different stakeholders within the supply chain and apportioning different roles to every contributor, traceability enables firms to solve issues emerging within the production chain effectively. Further, technologies like blockchain make it possible to record all transactions and movements of products transparently and immutably, thus giving a complete view of the production process. This proactive step in monitoring every stage of production ensures that at whatever point deviation from expected standards may occur, it can be timely identified and corrected to uphold efficiency and the integrity of the product.

5.4.2 Identification of the origin of raw materials

Other than monitoring production, traceability is essential in identifying the origin of raw materials. By tracing the movement of materials from source to finished product, it becomes possible for a company to ensure there is transparency and accountability in the supply chains. By using certifications such as BCI, GOTS, Good Earth CottonTM, or Regeneagri, companies can set requirements in sourcing raw materials for factors like water conservation, biodiversity, and land revitalization. It is from the understanding of the origin of raw materials and their traceability that ensures ethical practices are upheld, such as non-toxic farming, breeding, and harvesting processes. Transparency not only builds the trust of consumers but also drives the commitment of industry players towards sustainability and ethical practices. Integrating cutting-edge technologies such as artificial intelligence, traceability in companies such as Oritain identifies the geographical origin of raw materials with precision and thus promotes transparency

and sustainability in supply chains.

6 Addressing Sustainability Challenges in the Fashion Industry: AI for Zero-Waste Fashion

6.1 Responsible Optimization and Sustainability

6.1.1 Choice of Raw Materials

The primary advantage of AI is its ability to sort and analyze vast amounts of data in a very short time. When choosing textile materials, we primarily consider the origin of the material, its manufacturing process, its quality, as well as its environmental impact and sustainability. Indeed, so-called sustainable materials are increasingly sought after in the textile industry.

However, assessing the sustainability of a material is complex, which is where AI comes into play. For example, the startup "Higg Co" has developed a platform called the "Higg Materials Sustainability Index (MSI)" that uses artificial intelligence algorithms to assess the sustainability of raw materials used in the textile industry. This platform analyzes various environmental, social, and economic factors to assign sustainability scores to different raw materials, such as cotton, polyester, linen, and more. Well-known retail brands like H and M. have used this platform By relying on the data provided by Higg MSI, these brands have been able to compare and select raw materials with the least environmental impact throughout their lifecycle, from production to end-of-life disposal [27].

6.1.2 Identification of Recyclable Textiles

The textile industry is the third most polluting industry in the world. Indeed, the textile industry is estimated to be responsible for 20 percents of global water pollution as a result of dyeing and finishing during production. In addition to pollution, the textile industry consumes a huge amount of water and depletes the natural resources available on Earth. For exemple, to make just one cotton T-shirt, it takes approximately 2720 liters of water equivalent to what one adult would drink in three years [7]. So, despite choosing sustainable materials, a garment remains highly polluting even at the end of its life. It is therefore necessary to recycle textiles.
However, currently, it is impossible to recycle 100 percent of textile production. It is necessary to sort through and identify textiles that can be recycled and those that cannot, to maximize the recycling of those that can by giving them a second life. AI algorithms have been used to develop computer vision systems capable of automatically recognizing and classifying different types of textiles based on their material, fiber composition, texture, and strength.

These systems are integrated into automated sorting machines that separate recyclable textiles from non-recyclable waste quickly and accurately.
Additionally, AI can be used to identify the fibres composing a textile, providing essential information about whether it is recyclable and directing it to the appropriate recycling stream. Synthetic fibers are not recycled in the same way as natural fibers because they do not have the same intrinsic properties. Synthetic fibers are produced from non-renewable resources like petroleum. Recycling these fibers is done mechanically: they are melted and then transformed into pellets. These pellets can then either be spun into yarn or used for other applications. Natural fibers, on the other hand, come from plant sources like cotton and linen or animal sources like wool. To recycle these fibers, textiles are cut, shredded, and defibered [19]. Recycled natural fibers are typically reused in non-woven materials. It should be noted that recycled fiber loses mechanical properties: it will be less resistant, and lose tenacity, and elasticity.
Furthermore, artificial intelligence can analyze large amounts of data on recyclable textiles to continuously improve their classification and identification capabilities. These systems can learn to recognize new materials and adapt to variable conditions, optimizing the performance of the sorting process over time. AI thus contributes to improving recycling practices and reducing the environmental impact of the textile industry through innovative solutions [35].

6.1.3 Valorization of Materials

The integration of AI in the development of new materials constitutes another advancement in the textile industry. When developing new materials, material modeling enables accurate predictions of textile properties. These predictions form a substantial foundation for choosing the right material for each specific application. For example, the company Airbus uses AI in its creation process. Based on AI predictions, Airbus selects the appropriate material for each part of the aircraft [3]. This way, Airbus enhances the performance, durability, lightness, and efficiency of its products. This has allowed the company to remain competitive in the global aviation market while contributing to the reduction of its environmental impact through more fuel-efficient and durable aircraft.
Furthermore, AI can predict the behavior of these materials under different environmental conditions or even foresee potential malfunctions, thereby improving the reliability of the final product. Artificial intelligence can also adjust any qualitative parameters, such as temperature or production speed, to maximize the performance ratio of the entire production chain.
Consequently, material processing companies can significantly increase their yield and efficiency by optimizing the quality and precision of their opera-

tions while minimizing excess resources and waste. The use of AI enables companies to promote the development of more sustainable products and progress towards more environmentally friendly practices in their supply chain, whether textile or otherwise. It also allows them to communicate transparently with consumers about the sustainability of their products, thus responding to a growing demand for more responsible fashion.

6.2 Maximum reduction of textile waste

6.2.1 Circular Design to Minimize Losses

Circular design is a concept that reimagines how products are designed, manufactured, and used to minimize waste, extend the lifespan of products, and promote more efficient resource utilization. Unlike the traditional linear model of "take, make, dispose of," where resources are extracted, transformed into products, and ultimately disposed of at the end of their life, the circular design aims to create a system where materials and products are continuously reused, recycled, or reintegrated into the economy [25].
In the textile industry, a significant amount of raw material is lost during multiple stages of production. This loss occurs throughout the production chain. For example, many prototypes are made before production begins, resulting in waste. There are also defective pieces that do not pass quality control (with holes, size issues, broken threads, etc.). However, the stage where the most textile losses occur is during cutting: the shapes of the items are cut from fabric rolls. Patterns vary greatly depending on clothing models and sizes, leading to numerous material losses. To address this issue, many companies are turning to artificial intelligence to optimize the cutting stage [34].

Algorithms is automatically generating the optimized cutting patterns these take into account variations in size fabrics traits and limitations of manufacturing for minimizing waste of fabrics and to reduce losses.
Moreover AI system continue monitoring performance for textile cut systems and adjust parameters automatically real time optimise usage of materials. For instance when a defect's spotted in material, AI rearrange them cutting patterns to minimize loss around areas affected.
An additional major step towards circular designs are virtual modeling this relies on artificial intelligence simulate cutting process thus reducing losses enhance efficiency on production operations.

6.2.2 Anticipating trends to minimize waste

The textile industry generates a tremendous amount of waste during the production stage. It's more than 15 percent of waste generated even before the

item leaves the factory. Several companies are opting to use artificial intelligence to address this issue. Indeed, machine learning algorithms are capable of analyzing past trends and consumer purchasing behaviors to predict with the utmost accuracy the future demand for upcoming textile products. This allows manufacturers to produce the exact quantity of items, thereby reducing production surpluses and associated waste.

Moreover, AI could be used to optimize production processes in real-time. AI could adjust the production parameters on its own to minimize waste and maximize resource usage. Hence, AI systems could continuously monitor the performance of produced equipment and predict possible failure. In this way, companies would have time to intervene and prevent production stops and losses.
Furthermore, AI systems could optimize logistics and inventory management throughout the supply chain. This way, they could minimize delays and disruptions and optimize material ratio and storage capacity. For example, AI could monitor the real-time environmental impact of the production of textiles. This way, it could detect not only areas where the out- puts might fail standards, but also areas where they could be improved.
By combining all these approaches, textile companies can not only reduce their environmental footprint and costs but also contribute to a more sustainable and ethical fashion sector.

7 Strategy and Marketing Aspect of AI

7.1 State-of-the-art customer service

7.1.1 Presentation of AI-based customer support software

Artificial intelligence is known to be effective in tasks where humans are slower. Having customer service driven by this technology is a powerful way for companies to save time and increase efficiency. [38] [17]

Indeed, several AI-based customer support software applications are being used to improve the efficiency and quality of customer interactions. Software such as Zendesk, Intercom or even Chatbots can be used to automatically answer questions asked by customers, using frequently asked questions and redirecting the customer to a human in the event of a complex question. The Freshdesk platform can perform repetitive customer service tasks and improve ticket office management. There are many other software packages of this kind with more or less the same function.

It is very helpful for humans who work in customer service and customers

who no longer have to wait for a fast answer.

7.1.2 24/7 efficiency and availability

This means that the software can operate 24 hours a day, 7 days a week, unlike a human being who has a fixed number of working hours per day and per week. Artificial intelligence will therefore be able to help customers answer their questions, even when staff are not working. This will reduce staff workloads and enable customers to get instant answers. In this way, efficiency and speed of work can be maximized. [9]

7.1.3 Quick and accurate responses

The answers provided by the AI will be instantaneous, and sometimes more transparent than those provided by a human. A series of frequently asked questions is entered into the AI database, along with the answers that must be provided. If the request is more complex than expected, human assistance will be required. So in this case, the job is not replaced by the IA, but it's a great help for them. [9]

7.2 Market analysis and positioning

7.2.1 Market study to identify competitors and stand out

To function, artificial intelligence is equipped with a vast database. This includes much information about the fashion industry market, thanks to customer reviews left on the internet, trends, social networks, etc. The AI will make it easier and, above all, faster to sort through this information. In addition, predictive models can be used to anticipate future market trends. [16]

To study the competition in the market more closely, AI will be able to examine and collect data provided by the web, customers, and companies' key figures, to know exactly what their strengths and weaknesses are. This makes it easier to stand out from the crowd after an analysis that is as qualitative as quantitative. [12]

7.2.2 Pricing strategy

When carrying out an accurate market analysis, it is essential to focus on the prices set by other companies to define your prices.

Moreover, AI makes it possible to process the data on the market prices, demand elasticities, and consumer buying behavior, and, accordingly, AI can adjust prices dynamically with fluctuations in demand, depending on alternative prices of substitute goods, and with an eye to general market conditions and the company's business imperatives. [20]

As a result, artificial intelligence will decrease companies' problems with constructing prices and make it more flexible in case the market regulation alters. [12]

7.2.3 Search for partnerships

A good market study includes a reasonable search for partners. It can be time-consuming and difficult to find contacts to work with. The use of artificial intelligence makes this task much more straightforward. Its algorithms will gather as much data as possible on companies (sales sectors, financial performance, etc.) to identify the best potential partners. Moreover, it is possible to ask the AI directly for the criteria you are looking for in companies, to save even more time.

This technology will help you find the best partners, knowing as much about them as possible and avoiding working with companies with more risks than opportunities. In addition, AI will be able to continuously monitor partner performance, providing predictive analyses and success indicators. [12]

7.3 Marketing campaigns

7.3.1 Creation of fictional models for photos with AI

Spanish brand Stradivarius caused a sensation at Paris Fashion Week on 28 April 2023. The old-fashioned mannequins were replaced by entirely computer-generated models made by lIA. Creative director Isabella Rizzo said the aim was to make fashion more sustainable by combining creativity and technology.

They used 3D scanning to obtain as many different body shapes as possible so that the models looked real. Each model has been designed to act like a human being, all with different personalities thanks to their expressions, and they all pause in their walk on the catwalk.

It's a risk that Stradivarius has taken by breaking with the classic codes of catwalk shows and bringing new technologies to the fore. [31]

7.3.2 Virtual Clothing Try-on

Thanks to artificial intelligence, brands are developing new ideal technologies for their shoppers.

For several years now are introduced a revolutionary shopping experience by launching virtual try-on clothes assisted by artificial intelligence before you buy. The technology aims to solve one of the main challenges of online shopping: uncertainty about the size and fit of clothes. Customers can even personalize their virtual avatar based on their height, body shape, and even skin tone, for an even more personalized fitting experience.

Luxury brand InnovFashion has similarly introduced virtual fittings using photos that customers can import themselves. However, the brand offers customers the chance to see how the fabrics react to their movements, simulating the actual wearing of the garments. Many brands are gradually adopting this type of technology, making it easier for customers to shop online.

Using artificial intelligence has many advantages for both businesses and customers. However, it is essential to set limits to its use to avoid the potential risks it could have, particularly in terms of marketing and creativity.

7.4 Potential risks of AI in fashion marketing and creativity

7.4.1 Counterfeiting and Plagiarism

When using new technologies such as artificial intelligence, certain risks are to be expected. One of the first drawbacks is plagiarism and counterfeiting.

By asking a program to design a new collection in line with current trends, there is a risk that the clothes will be similar between brands, leading to the risk of counterfeiting.
Luxury brand Hermès has accused artist Mason Rothschild of plagiarising its famous Birkin bag. Using artificial intelligence, he had developed a bag under the name MetaBirkins that earned him over a million dollars. It was ruled that the artist had profited from Hermès products and lost the debate. [11]

Virtual clothing trials make it easier for malicious people to imitate the look of a brand's clothing and its effect on humans.

For example, major retailers such as Shein use artificial intelligence to find their designs and create their clothes. They take their cues from many

brands, both well-known and lesser-known. As a result, some small independent brands that design their own collections and have to set higher prices to be profitable are competing with such brands, selling their products at low cost. [15]

Artificial intelligence can also use designs, logos, or models protected by copyright or patents. If it is used, the brand is infringing without even realizing it. Extreme care must therefore be taken when using AI to create designs.

7.4.2 Advertising and its limits

Generally used for industrial process optimization or sales and customer experience, artificial bits of intelligence have tended to remain on the sidelines of the creative process. However, the emergence of digitalization and its technologies during the COVID pandemic has forced brands to rethink their means of communication and their marketing strategy.

AI is developing and invading the fashion and luxury industries, revolutionizing the world of advertising and inspiring creativity and innovation.

The video creator Ian Padgham is well known for his large-scale advertising for major brands such as Jacquemus and Maybelline. For the release of the Bambino bag, the brand called on Padgham for a masterful advert. He posted a viral video of Bambino XXL bags in different colors rolling through the streets of Paris. [21] For a new Maybelline mascara campaign, he also posted a video of the mascara brush hanging on the walls of the metro station in London, which brushes the eyelashes hanging above the metro when it reaches the station. [23]

Of course, even though these adverts are made using artificial intelligence, they look real and have enabled the brands to mount a high-profile marketing campaign.
However, it can sometimes be difficult to tell the difference between the real and virtual worlds, leading to several reactions, some of them harmful.

Some well-known companies, such as Calvin Klein and Fenty Beauty, Rihanna's cosmetics line, have employed Lil Miquela and Shudu Gram, fictional characters created by artificial intelligence, in their marketing campaigns. These two 'women' have gained a large following on social networks and are considered to be influential personalities.
Customers' main concerns are honesty and authenticity. The use of these characters can be misleading and deceive consumers if they discover that

they are not real people.
In addition, these characters are designed to meet certain beauty standards, which can have a negative impact on the way people perceive natural beauty and lead them to feel insecure about their own appearance. [14]

Some fashion industry professionals also point out that the use of virtual influencers could potentially threaten employment opportunities for accurate models, especially in a sector where competition for advertising contracts is already intense. [11]

Finally, brands expose themselves to risks by using artificial intelligence for their marketing. Indeed, they may be subject to legal proceedings if it is not transparent that AI has been used to imply that advertising is misleading, if there are marks of cultural appropriation that could be disturbing and even concerning the image rights of fictional characters.

So, artificial intelligence is a very good medium and can offer many opportunities, both in marketing and across the supply chain, but it is essential to set limits on its use so as not to affect workers, consumers, and younger audiences.

8 Case study

Throughout this project we've been talking about the power of artificial intelligence in the world of fashion. Let's now focus on Chat GPT and its many functions, which are helping to improve this industry.

Chat GPT has a wealth of marketing expertise. It is capable of analysing consumer preferences in order to help brands tailor their future collections to their target audience.
It even enables brands to predict trends over the coming years.
We decided to ask him some more or less specific questions, based on company criteria, and this is what we got from him.

An example of a table comparing women's fashion preferences according to age :

Criteria	Women aged 20 to 30	Women aged 40 to 50	Women aged 60 to 70
Fashion Style	Inclination towards adopting the latest fashion trends	Preference for elegant and timeless clothing	Prefers a classic and comfortable style
Preferred Colors	Openness to a wide range of bright and bold colors	Often favors neutral and sober colors	Appreciates soft and soothing colors
Comfort	Expectation of a balance between style and comfort	Importance placed on comfort without sacrificing style	Priority on comfort without compromising on elegance
Technology Usage	Familiarity with digital trends and social media	Moderate use of technology	Less comfortable with the latest technologies
Preferred Activities	Seeks social experiences and adventures	Prefers family activities and quiet leisure	Interest in calm activities and creative hobbies
Accessories	Enthusiastic about trendy and bold accessories	Preference for classic and timeless accessories	Appreciates simple and elegant accessories

An example of a table comparing the preferences of women in fashion according to their salary income:

Criteria	**Low Incomes**	**Medium Incomes**	**High Incomes**	**Very High Incomes**
Fashion Style	Seeks affordable and practical items	Inclination to invest in quality pieces	Preference for luxury brands and high-end items	Pursuit of luxury items and prestigious designers
Preferred Brands	Fast fashion brands and affordable items	Variety of brands and price ranges	Luxury brands and designers	Exclusive luxury brands and haute couture
Shopping Frequencies	Occasional purchases and bargain hunting	Frequent purchases of quality clothing	Regular purchases of high-end clothing	Frequent purchases of haute couture pieces and exclusive collections
Materials and Finishes	Preference for affordable materials	Price sensitivity, but seeks quality	Preference for high-quality materials and luxurious finishes	Pursuit of the highest quality materials and finishes
Accessories	Seeks affordable and trendy accessories	Prefers quality accessories	Preference for branded accessories	Seeks luxury and designer accessories
Fashion Preferences	Inclination to follow the latest fashion trends	Interest in timeless pieces and quality basics	Interest in the latest fashion trends and unique pieces	Pursuit of haute couture pieces and renowned designers

A table illustrating the criteria on which ready-to-wear brands base their new collections, with examples for each criterion:

Criteria	2023	2024	2025	2026
Trends in Fashion	- Trend towards pastel tones: Soft and soothing colors such as blush pink and mint green. - Style: Chic minimalism with clean lines and fluid silhouettes.	- Color of the year: "Luminous Aurora" (Pantone 13-6022) announced for 2024. - Style: Return of floral patterns and bold prints.	- Color of the year: "Luminous Aurora" (Pantone 13-6022) announced for 2024. - Style: Return of floral patterns and bold prints.	- Neon color trend: Bright and vibrant colors for a bold and energetic aesthetic. - Style: Reinterpretation of the 80s with tailored cuts and punk-rock elements.
Analysis of Previous Sales	- Cargo pants were in high demand in 2022. - Long coats saw a significant increase in sales.	- Jumpsuits were the best-selling products in 2023. - Accessories such as backpacks recorded an increase in sales.	- Bow blouses were popular in 2024. - Platform sneakers dominated the shoe market.	- Vegan leather jackets were a staple of the season. - Chunky heeled boots were a popular choice among consumers.
Market Study	- The loungewear clothing market continues to grow with increased demand for comfortable and versatile clothing. - Consumers are sensitive to environmental impact and seek brands committed to sustainability.	- Demand for party and special occasion clothing increased with the reopening of social events. - Consumers increasingly prioritize the online shopping experience, including customization and ease of navigation.	- Consumer trends are shifting towards a more functional and sustainable fashion. - Consumers seek versatile clothing that can be worn in different contexts.	- Demand for performance sportswear is increasing with the growing popularity of fitness and outdoor activities. - Consumers seek ethical and eco-friendly fashion options.

Criteria	2023	2024	2025	2026
Competition Analysis	- A competing brand launched a bold advertising campaign highlighting body diversity. - Another brand opened a new flagship store with an immersive customer experience.	-A rival brand has introduced a range of workwear adapted to the new hybrid office standards.- Another brand has developed a mobile application offering augmented reality shopping.	- A competing brand launched a limited edition capsule collection with recycled materials. - Another brand adopted a transparent approach to the supply chain by allowing consumers to track the origin of their clothing.	- A competing brand launched a marketing campaign focused on inclusivity and diversity. - Another brand developed an omnichannel e-commerce strategy to enhance the customer experience.
Seasonal Considera-tion	- Spring/ Summer Collection: Linen jumpsuits, midi skirts, platform sandals. - Fall/Winter Collection: Quilted jackets, cashmere sweaters, high boots.	- Spring/ Summer Collection: Off-shoulder blouses, wide-leg pants, rope espadrilles. - Fall/Winter Collection: Structured blazers, velvet dresses.	- Spring/ Summer Collection: Animal print maxi dresses, silk shirts, square-toe sandals. - Fall/Winter Collection: Faux fur coats, turtleneck sweaters, pointed-toe boots.	- Spring/ Summer Collection: Linen jackets, leather shorts, platform sandals. - Fall/Winter Collection: Recycled wool coats, ethical cashmere sweaters, vegan leather boots.

Criteria	2023	2024	2025	2026
Innovation and Creativity	- Integration of blockchain traceability technology to ensure transparency in product supply chain.- Development of an innovative sportswear line with breathable fabrics and antibacterial properties.	- Use of artificial intelligence to personalize product recommendations based on customer preferences and purchase history. - Creation of a range of modular and interchangeable clothing to promote sustainability and versatility.	- Launch of a collection of unisex clothing designed to fit a variety of body types and styles. - Introduction of low-impact dyeing technology to reduce the environmental footprint of products.	- Collaboration with emerging designers to create exclusive collections and promote new talent. - Use of innovative materials such as mushroom leather and recycled plastic bottle fabric for sustainable and eco-friendly products.

When it comes to advertising marketing, Chat GPT can do anything you ask of it. For example, we asked him to create a post on social networks to promote a new swimming costume.

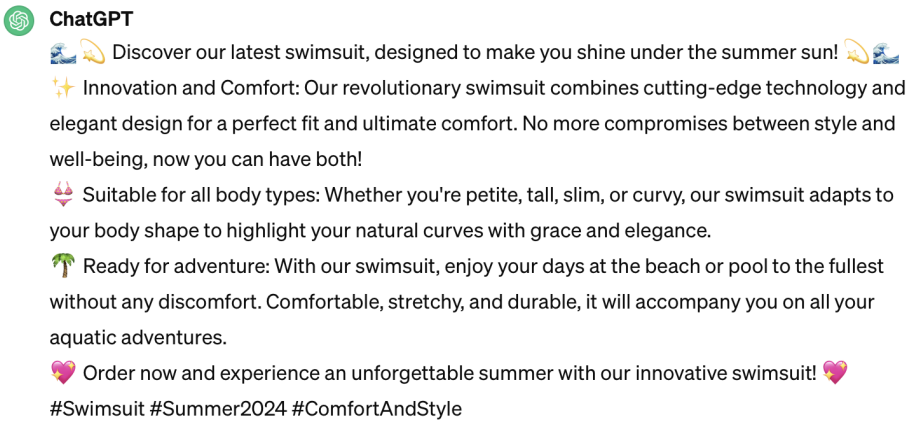


Figure 1: Instagram post for a new swimwear by Chat GPT

With the version 4 of Chat GPT, we can ask what we want and the AI can transcribe it into photo or video. That is what a lot of companies are doing now for their publicity. For example, we asked him to create a collection of 5 garments with a little description and generate it as a picture:

Figure 2: A clothing collection by Chat GPT

1. Asymmetrical dress: Use of fabrics that change with the light, with asymmetrical with asymmetrical cuts and colours such as bright turquoise and coral. and coral.

2. Fluid jumpsuit: One-piece jumpsuit with abstract floral patterns incorporating conductive threads to create a subtle luminous effect.

3. Coordinating top and shorts: Together with dynamic digital prints and colours such as neon yellow and electric pink, designed to reflect the to reflect the energy of summer.

4. Light jacket: Jacket in breathable fabric, dyed lime green and ocean blue, with reflective elements for night-time visibility.

5. Long pleated skirt: Skirt in a gradient of colours from deep purple to light blue using a fabric that reacts to temperature to change colour with heat. change colour with heat.

We asked it after to describe the way it was manufactured (weaving, knitting, materials...). Here is the answer:

1 Asymmetric dress: For a fabric that changes according to the light, you could consider synthetic fibres such as polyester or nylon, which can be can be mixed with reflective yarns to create the colour-changing effect. colour change effect. For contrasting panels, lightweight, flowing fabrics such as chiffon or crepe could be used to ensure an elegant drape. elegant drape.

2. Fluid combination: Synthetic fibres such as polyester or spandex would be ideal for creating a fluid, stretchy fabric, perfect for a comfortable comfortable suit. Incorporating conductive yarns into the fabric would require the use of conductive fibres such as metallic wire or carbon thread.

3. Coordinating top and shorts: For vibrant digital prints, polyester or cotton/polyester blend cotton/polyester blend fabrics would provide a solid base for accurate colour accurate colour reproduction. For comfort and breathability, a cotton blend stretch fabric with spandex could be used for the top and shorts.

4. Light jacket: For a lightweight, breathable jacket, natural fibres such as cotton or linen could be used. cotton or linen could be used, possibly mixed with synthetic fibres to add synthetic fibres to add durability and water resistance. resistance. Reflective threads could be integrated into the fabric using special reflective polyester yarns.

5. Pleated long skirt: Synthetic fibres such as polyester or nylon would be ideal to create a lightweight and durable pleated skirt. For temperature-reactive colour shading, thermochromic fibres could be fibres could be blended into the fabric to create the colour change effect. colour change effect.

As consumers, using GPT Chat can be very useful. You can ask it lots of questions about what you want and what you're looking for, to make things quicker and easier. For example, I'm a tall girl and I'd like to buy linen trousers in my size. I can then ask her for advice on brands, the size to choose according to my measurements, etc. Here's a table summarising

the requests for linen trousers according to age, size and other criteria:

Age	**Size (French)**	**Other criteria**	**Suggested brands**
20 years old	38	Tall (1 meter 80 centimeters)	Zara, ASOS tall, H and M tall, Mango, Gap
40 years old	40	Small (1 meter 50 centimeters)	Topshop small, Petite studio, Ann Taylor small, J.Crew small
50 years old	38	Middle (1 meter 65 centimeters)	Massimo Dutti, Linenfox, COS, Reformation, Land's End

However, the process for the AI is not so easy than we thought. In fact, Chat GPT needs a lot of upstream programming to help us and answer our questions ans our requests. In fact, prompt engineering is essential for working with Chat GPT. When users are providing more accurate, relevant and helpful responses by using the prompt, it helps improve the communication between the AI and the users. We did some prompt that can be used for some things: [28]

Here are kinds of Pyhton programs that need to be executed in order to avoid counterfeiting :

Here is an example of a clothing recommendation system using Python and the scikit-learn library to implement a collaborative filtering algorithm called "user-based filtering". In this example a dummy dataset containing user ratings of clothes is used.

```python
import pandas as pd
import numpy as np
from sklearn.metrics.pairwise import cosine_similarity

# Create a dummy dataset
data = {
    'User': ['Alice', 'Alice', 'Bob', 'Bob', 'Bob', 'Charlie', 'Charlie'],
    'Clothing': ['Shirt', 'Dress', 'Shirt', 'Pants', 'Skirt', 'Dress', 'Pants'],
    'Rating': [5, 4, 3, 5, 2, 4, 3]
}
df = pd.DataFrame(data)

# Transform the dataset into a user-item matrix
user_item_matrix = df.pivot_table(index='User', columns='Clothing', values='Rating', fill_value=0)

# Calculate cosine similarity between users
user_similarity = cosine_similarity(user_item_matrix)

# Define a function to get recommendations for a given user
def get_recommendations(user, user_similarity_matrix, user_item_matrix):
    similar_users = user_similarity_matrix[user_item_matrix.index == user]  # Select similarity row for the given user
    user_ratings = user_item_matrix.loc[user]  # Get ratings of the given user

    # Weighted sum of ratings from similar users
    weighted_sum = np.dot(similar_users, user_item_matrix.values)
    weighted_sum = weighted_sum / similar_users.sum(axis=1)[:, None]  # Normalize by sum of similarities

    # Exclude items that the user has already rated
    recommendations = pd.DataFrame(weighted_sum, columns=user_item_matrix.columns, index=[user])
    recommendations = recommendations.drop(user_ratings.index, axis=1)

    # Sort recommendations by descending rating
    recommendations = recommendations.T.squeeze().sort_values(ascending=False)

```

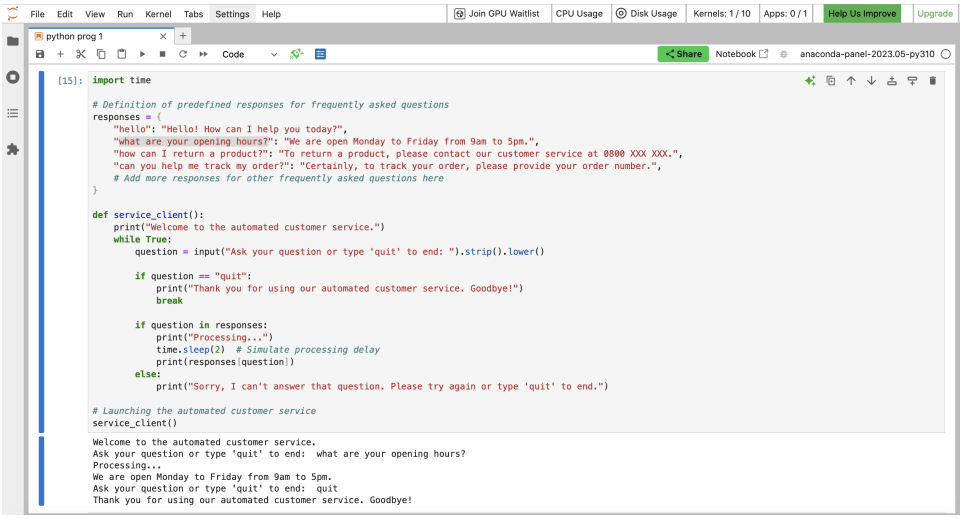
Figure 3: Python user-based filtering algorithm (1)

```python

    # Sort recommendations by descending rating
    recommendations = recommendations.T.squeeze().sort_values(ascending=False)

    return recommendations

# Get recommendations for the user 'Alice'
recommendations = get_recommendations('Alice', user_similarity, user_item_matrix)
print(recommendations)
```

```
Series([], Name: Alice, dtype: float64)
```

Figure 4: Python user-based filtering algorithm (2)

We mentioned above the use of artificial intelligence for automated customer service. Chat GPT is capable of answering questions asked by consumers, by entering a database of typical questions and their associated answers. If the buyer is unable to answer the question, he or she will automatically be redirected to a human. As we said, this means that customer service is available 24 hours a day, 7 days a week, faster and more efficiently. Here is a simple example of the Python code used for this type of service:



```
[15]: import time

# Definition of predefined responses for frequently asked questions
responses = {
    "hello": "Hello! How can I help you today?",
    "what are your opening hours?": "We are open Monday to Friday from 9am to 5pm.",
    "how can I return a product?": "To return a product, please contact our customer service at 0800 XXX XXX.",
    "can you help me track my order?": "Certainly, to track your order, please provide your order number.",
    # Add more responses for other frequently asked questions here
}

def service_client():
    print("Welcome to the automated customer service.")
    while True:
        question = input("Ask your question or type 'quit' to end: ").strip().lower()

        if question == "quit":
            print("Thank you for using our automated customer service. Goodbye!")
            break

        if question in responses:
            print("Processing...")
            time.sleep(2)  # Simulate processing delay
            print(responses[question])
        else:
            print("Sorry, I can't answer that question. Please try again or type 'quit' to end.")

# Launching the automated customer service
service_client()

Welcome to the automated customer service.
Ask your question or type 'quit' to end:  what are your opening hours?
Processing...
We are open Monday to Friday from 9am to 5pm.
Ask your question or type 'quit' to end:  quit
Thank you for using our automated customer service. Goodbye!
```

Figure 5: Automatic customer service by GPT Chat

The program allows us to ask questions and get answers (or not) to our questions. This is how AI-controlled customer services are programmed. Of course, the program here is very simplified.

9 Conclusion

The fashion industry is one of the oldest human activities and has evolved over the centuries by constantly adapting to technological and societal advancements. This industry faces numerous challenges, such as short product life cycles, extended supply chain lead times, and highly fluctuating demand, making it one of the most complex environments to manage.

Integrating artificial intelligence (AI) into the fashion industry offers a promising avenue for addressing some of these challenges. AI can streamline processes, optimize supply chain management, and enhance customer experiences through personalized recommendations and targeted marketing strategies. By leveraging AI, fashion brands can gain insights into consumer preferences, forecast trends more accurately, and improve operational efficiency.

AI can offers significant advantages, particularly in the realm of marketing. AI-driven analytics can provide invaluable insights into consumer behavior, enabling fashion brands to develop more targeted and effective marketing campaigns. By leveraging AI-powered tools for customer segmentation, personalized messaging, and content optimization, fashion companies

can enhance their marketing strategies and improve their competitive edge in the market.

Nevertheless, it's important to recognize that AI is not a panacea for all the industry's woes. There are limitations to its capabilities, particularly in areas that require human creativity, intuition, and cultural understanding. While AI can augment decision-making processes, it cannot entirely replace the human touch in design, branding, and customer engagement.

Moreover, the integration of AI in marketing also presents challenges. Automated decision-making algorithms may perpetuate biases and inequalities, leading to unintended consequences such as reinforcing stereotypes or excluding certain demographic groups. Moreover, the reliance on AI for targeted advertising raises concerns about consumer privacy and data security, especially in light of recent controversies surrounding data breaches and misuse.

Furthermore, the widespread adoption of AI in the fashion industry raises ethical and social concerns, such as data privacy, algorithmic bias, and the potential displacement of human labor. Balancing technological innovation with ethical considerations is paramount to ensure that AI-driven advancements benefit both businesses and society as a whole.

In conclusion, while AI holds immense promise for revolutionizing the fashion industry, its integration must be approached thoughtfully and ethically. By leveraging the strengths of AI while acknowledging its limitations, fashion companies can navigate the complexities of the industry more effectively and sustainably, ultimately driving innovation, enhancing the consumer experience, promoting environmental stewardship, and addressing negative impacts in marketing practices.

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